

$$\Rightarrow \int_0^1 \int_0^{y^2} \left[\frac{x^{-2}}{2} \right]_1^{z+1} dz dy$$

$$\Rightarrow \int_0^1 \int_0^{y^2} \left[\frac{-(z+1)^{-2}}{2} \right]_1^{z+1} dz dy$$

$$\Rightarrow \int_0^1 \int_0^{y^2} \left[\frac{1}{2(z+1)} + \frac{1}{2} \right] dz dy$$

$$\Rightarrow \int_0^1 \left[\frac{1}{2(z+1)} + \frac{1}{2} z \right]_0^{y^2} dy$$

$$\Rightarrow \int_0^1 \left[\frac{1}{2(z+1)} + \frac{1}{2} z \right]_0^{y^2} dy$$

$$\Rightarrow \int_0^1 \left[\frac{1}{2} (y^2+1) + \frac{1}{2} (y^2) \right]_0^1 dy$$

$$\Rightarrow \int_0^1 \left[\frac{1}{2} (y^2+1) + \frac{1}{2} y^2 - \frac{1}{2} \right] dy$$

$$\Rightarrow \left[\frac{1}{2} \tan^{-1}(y) + \frac{1}{2} y^3 - \frac{1}{2} y \right]_0^1$$

$$\Rightarrow \left[\frac{1}{2} \tan^{-1}(1) + \frac{1}{2} (1)^3 - \frac{1}{2} (1) \right]_0^1$$

$$\left[0 + 0 - 0 \right]$$

$$\Rightarrow \left[\frac{1}{2} \cdot \frac{1}{4} \pi + \frac{1}{6} - \frac{1}{2} \right]$$

$$\Rightarrow \frac{\pi}{8} + \frac{1-3}{6}$$

$$\Rightarrow \frac{\pi}{8} + \left(\frac{-2}{6} \right)$$

$$\Rightarrow \left[\frac{\pi}{8} - \frac{1}{3} \right] \text{ Ans}$$

$$\Rightarrow \int_0^1 \int_y^1 (ye^x - y) dx dy$$

$$\Rightarrow \int_0^1 \int_y^1 ye^x dx dy - \int_0^1 \int_y^1 y dx dy$$

$$\Rightarrow \int_0^1 [ye^x]_y^1 dy - \int_0^1 [yx]_y^1 dy$$

$$\Rightarrow \int_0^1 (ye^1 - ye^y) dy - \int_0^1 (y - y^2) dy$$

$$\Rightarrow \left[\frac{y^2 e}{2} \right]_0^1 - \int_0^1 ye^y dy - \left[\frac{y^2}{2} - \frac{y^3}{3} \right]_0^1$$

$$\Rightarrow \left[\frac{y^2 e}{2} \right]_0^1 - \left[y \int e^y dy - \int \frac{d}{dy}(y) \int e^y dy dy \right] - \left[\frac{y^2}{2} - \frac{y^3}{3} \right]_0^1$$

$$\Rightarrow \left[\frac{y^2 e}{2} \right]_0^1 - \left[ye^y - \int (1) e^y dy \right] - \left[\frac{y^2}{2} - \frac{y^3}{3} \right]_0^1$$

$$\Rightarrow \left[\frac{y^2 e}{2} \right]_0^1 - \left[ye^y - e^y \right]_0^1 - \left[\frac{y^2}{2} - \frac{y^3}{3} \right]_0^1$$

$$\Rightarrow \left[\frac{(1)^2 e}{2} - 0 \right] - \left[(1)(e^1) - e^0 \right] - \left[-1 \right] - \left[\frac{1}{2} - \frac{1}{3} \right]$$

$$\Rightarrow \left(\frac{e}{2} \right) - [1] - [-1] - \left[\frac{1}{2} - \frac{1}{3} \right]$$

$$= \frac{e}{2} - 1 - \frac{1}{6}$$

$$\Rightarrow \frac{1}{2} (e - \frac{1}{6} - 1)$$

$$\Rightarrow \frac{1}{2} e - \frac{1-6}{6}$$

$$\Rightarrow \frac{1}{2} e - \frac{7}{6} \text{ Ans}$$

15. $\iiint_E (1/x^3) dV$, where

$$E = \{(x, y, z) \mid 0 \leq y \leq 1, 0 \leq z \leq y^2, 1 \leq x \leq 2\}$$

Sol

$$\Rightarrow \int_0^2 \int_0^1 \int_1^{y^2 z+1} (1/x^3) dx dz dy$$

$$\Rightarrow \int_0^2 \int_0^1 \int_1^{y^2 z+1} x^{-3} dx dz dy$$

$$13 - \iiint_E y \, dV$$

$$E = \{(x, y, z) \mid 0 \leq x \leq y, 0 \leq y \leq 1, 0 \leq z \leq y\}$$

Sol

$$\Rightarrow \int_0^1 \int_0^y \int_0^y y \, dz \, dx \, dy$$

$$= \int_0^1 y \, dy - \int_0^1 (y - y^2) \, dy$$

$$= \left[\frac{y^2}{2} \right]_0^1 - \left[\frac{y^2}{2} - \frac{y^3}{3} \right]_0^1$$

$$\Rightarrow \left[\frac{y^2}{2} \right]_0^1 - \left[y \int_0^y e^y \, dy - \int_0^1 \left(\frac{d}{dy} \int_0^y e^y \, dy \right) dy \right] - \left[\frac{y^2}{2} - \frac{y^3}{3} \right]_0^1$$

$$\Rightarrow \left[\frac{y^2}{2} \right]_0^1 - \left[y e^y - \int_0^1 (1) e^y \, dy \right] - \left[\frac{y^2}{2} - \frac{y^3}{3} \right]_0^1$$

$$\Rightarrow \left[\frac{y^2}{2} \right]_0^1 - \left[y e^y - e^y \right]_0^1 - \left[\frac{y^2}{2} - \frac{y^3}{3} \right]_0^1$$

$$\Rightarrow \left[\frac{(1)^2}{2} - 0 \right] - \left[(1)(e^1) - e^0 \right] - \left[-1 \right] - \left[\frac{1}{2} - \frac{1}{3} \right]$$

$$\Rightarrow \left(\frac{1}{2} \right) - [1] - \left[\frac{3-2}{6} \right]$$

$$= \frac{1}{2} - 1 - \frac{1}{6}$$

$$\Rightarrow \frac{1}{2} \left(\frac{3}{3} - \frac{6}{6} - \frac{1}{6} \right)$$

$$= \frac{1}{2} \left(\frac{3-6-1}{6} \right)$$

$$\Rightarrow \frac{1}{2} \left(\frac{-4}{6} \right) = -\frac{2}{6} = -\frac{1}{3}$$

$$15. \iiint_E (1/x^3) \, dV, \text{ where}$$

$$E = \{(x, y, z) \mid 0 \leq y \leq 1, 0 \leq z \leq y, 1 \leq x \leq 2\}$$

Sol

$$\Rightarrow \int_0^1 \int_0^y \int_1^{2+y} (1/x^3) \, dx \, dz \, dy$$

$$\Rightarrow \int_0^1 \int_0^y \int_1^{z+1} x^{-3} \, dx \, dz \, dy$$

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13- $\iiint_E y \, dV$ where

$$E = \{(x, y, z) \mid 0 \leq x \leq 3, 0 \leq y \leq x, x-y \leq z \leq x+y\}$$

Sol

$$\Rightarrow \int_0^3 \int_0^x \int_{x-y}^{x+y} y \, dz \, dy \, dx$$

$$\Rightarrow \int_0^3 \int_0^x \left[yz \right]_{x-y}^{x+y} dy \, dx$$

$$\Rightarrow \int_0^3 \int_0^x [y(x+y)] - [y(x-y)] dy \, dx$$

$$\Rightarrow \int_0^3 \int_0^x (xy + y^2 - xy + y^2) dy \, dx$$

$$\Rightarrow \int_0^3 \int_0^x 2y^2 dy \, dx$$

$$\Rightarrow \int_0^3 \left[\frac{2y^3}{3} \right]_0^x dx$$

$$\Rightarrow \int_0^3 \left[\frac{2(x)^3}{3} - 0 \right] dx$$

$$\Rightarrow \int_0^3 \frac{2x^3}{3} dx$$

$$\Rightarrow \left[\frac{2x^4}{12} \right]_0^3$$

$$\Rightarrow \left[\frac{(3)^4}{6} \right] - 0$$

$$\Rightarrow \frac{81}{6} = \left[\frac{27}{2} \right] \text{ Ans}$$

14. $\iiint_E e^{z/y} \, dV$ where

$$E = \{(x, y, z) \mid 0 \leq y \leq 1, y \leq x \leq 1, 0 \leq z \leq xy\}$$

Sol

$$\int_0^1 \int_y^1 \int_0^{xy} e^{z/y} dz \, dx \, dy$$

$$\Rightarrow \int_0^1 \int_y^1 \left[(y) e^{z/y} \right]_0^{xy} dx \, dy$$

$$\Rightarrow \int_0^1 \int_y^1 [(y) e^{xy/y}] - y(e)^{0/y} dx \, dy$$

$$\Rightarrow \int_0^1 \left(e^{1-x^2} - e^{2-x^2} \right) dx - \int_0^1 \frac{x}{2} dx$$

$$\Rightarrow \left(\frac{e^{1-x^2}}{-2x} + \frac{e^{2-x^2}}{2x} \right) dx - \int_0^1 \frac{x}{2} dx$$

$$\Rightarrow \int_0^1 -\frac{x}{2} e^{1-x^2} dx + \int_0^1 \frac{x}{2} e^{2-x^2} dx - \int_0^1 \frac{x}{2} dx$$

Substitution (1)

$$\begin{cases} u = 1-x^2 \\ du = -2x dx \\ \frac{du}{-2} = x dx \end{cases}$$

Substitution (2)

$$\begin{cases} u = 2-x^2 \\ du = -2x dx \\ \frac{du}{-2} = x dx \end{cases}$$

$$\Rightarrow \int_0^1 \frac{-1}{(-2)(2)} e^u du + \int_0^1 \left(\frac{1}{2} \right) \frac{e^u}{(-2)} du - \left[\frac{x^2}{2} \right]_0^1$$

$$\Rightarrow \int_0^1 \frac{1}{4} e^u du + \int_0^1 \frac{-1}{4} e^u du - \left[\frac{x^2}{2} \right]_0^1$$

$$\Rightarrow \frac{1}{4} [e^u]_0^1 + \left(\frac{-1}{4} \right) [e^u]_0^1 - \left[\frac{(1)^2}{2} \right] - 0$$

$$\Rightarrow \left[\frac{1}{4} e^{1-x^2} \right]_0^1 - \left[\frac{1}{4} e^{2-x^2} \right]_0^1 - \frac{1}{2}$$

$$\Rightarrow \frac{1}{4} \left\{ (e^{1-1} - e^{1-1}) - (e^{2-1} - e^{2-1}) \right\} - \frac{1}{2}$$

$$\Rightarrow \frac{1}{4} [1 - e^1 - e^1 + e^2 - 1]$$

$$\Rightarrow \boxed{\frac{1}{4} [e^2 - 2e^1]} \text{ Ans}$$

$$\Rightarrow \int_1^3 \frac{8}{3y} + \frac{1}{3y} + 2 - \frac{1}{2} dy$$

$$\Rightarrow \int_1^3 \frac{3}{y} + \frac{4-1}{2} dy$$

$$\Rightarrow \int_1^3 \frac{3}{y} + \frac{3}{2} dy$$

$$\Rightarrow \left[3 \ln y + \frac{3}{2} y \right]_1^3$$

$$\Rightarrow \left[3 \ln 3 + \frac{3}{2} (3) \right] - \left[3 \ln 1 + \frac{3}{2} (1) \right]$$

$$\Rightarrow 3 \ln 3 + \frac{9}{2} - \frac{3}{2}$$

$$\Rightarrow 3 \ln 3 + \frac{6}{2}$$

$$\Rightarrow 3 (\ln 3 + 1)$$

$$\Rightarrow \boxed{3 \ln 3 + 3} \text{ Ans}$$

$$8. \int_0^1 \int_0^1 \int_0^{2-x^2-y^2} xye^z dz dy dx$$

$$\text{Sol} \int_0^1 \int_0^1 \left[xye^z \right]_0^{2-x^2-y^2} dy dx$$

$$\Rightarrow \int_0^1 \int_0^1 \left[xye^{(2-x^2-y^2)} \right] - xye^{(0)} dy dx$$

$$\Rightarrow \int_0^1 \int_0^1 xye^{2-x^2-y^2} - xy dy dx$$

$$\Rightarrow \int_0^1 \int_0^1 xye^{2-x^2-y^2} dy dx - \int_0^1 \int_0^1 xy dy dx$$

$$\Rightarrow \begin{cases} u = 2-x^2-y^2 \\ du = -2y dy \\ \frac{du}{-2} = y dy \end{cases}$$

$$= \int_0^1 \int_0^1 -\frac{x}{2} e^u du dx - \int_0^1 \int_0^1 xy dy dx$$

$$\Rightarrow \frac{2}{2} \int_0^{\pi/2} \int_0^{2\pi} \sin(x+z) dz dx$$

$$\Rightarrow \int_0^{\pi/2} \left[-\cos(x+z) \right]_0^{2\pi} dx$$

$$\Rightarrow \int_0^{\pi/2} -\cos(x+2\pi) - (-\cos(x)) dx$$

$$\Rightarrow \int_0^{\pi/2} -\cos(3x) + \cos(x) dx$$

$$\Rightarrow \left[-\frac{\sin 3x}{3} + \sin x \right]_0^{\pi/2}$$

$$\Rightarrow -\left[\frac{\sin(3(\pi/2))}{3} - \sin(\pi/2) \right] = 0$$

$$\Rightarrow +\frac{1}{3} + 1$$

$$\Rightarrow \frac{1+3}{3}$$

$$\Rightarrow \frac{4}{3} \text{ Ans}$$

$$7. \int_1^3 \int_{-1}^2 \int_{-y}^3 \frac{z}{y} dx dz dy$$

$$\Rightarrow \int_1^3 \int_{-1}^2 \int_{-y}^3 \frac{z}{y} dx dz dy$$

$$\Rightarrow \int_1^3 \int_{-1}^2 \left[\frac{xz}{y} \right]_{-y}^3 dz dy$$

$$\Rightarrow \int_1^3 \int_{-1}^2 \left[\frac{(3)(z)}{y} \right] - \left[\frac{(-y)(z)}{y} \right] dz dy$$

$$\Rightarrow \int_1^3 \int_{-1}^2 \frac{z^2}{y} + z dz dy$$

$$\Rightarrow \int_1^3 \left[\frac{z^3}{3y} + \frac{z^2}{2} \right]_{-1}^2 dy$$

$$\Rightarrow \int_1^3 \left[\frac{(2)^3}{3y} + \frac{(2)^2}{2} \right] - \left[\frac{(-1)^3}{3y} + \frac{(-1)^2}{2} \right] dy$$

$$\Rightarrow \int_1^3 \left(\frac{8}{3y} + 2 \right) - \left(-\frac{1}{3y} + \frac{1}{2} \right) dy$$

$$\Rightarrow \int_1^2 \left(-2z + \frac{(2z)^2}{2} \right) - 0 \, dz$$

$$\Rightarrow \int_1^2 -2z + 2z^2 \, dz$$

$$\Rightarrow \left[\frac{-2z^2}{2} + \frac{2z^3}{3} \right]_1^2$$

$$\Rightarrow \left[-(2)^2 + \frac{2(2)^3}{3} \right] - \left[-(1)^2 + \frac{2(1)^3}{3} \right]$$

$$\Rightarrow \left[-4 + \frac{16}{3} \right] - \left[-1 + \frac{2}{3} \right]$$

$$\Rightarrow \left(\frac{-12 + 16}{3} \right) - \left(\frac{-3 + 2}{3} \right)$$

$$\Rightarrow \frac{4}{3} - \left(-\frac{1}{3} \right)$$

$$\Rightarrow \frac{4}{3} + \frac{1}{3} = \boxed{\frac{5}{3}} \text{ Ans}$$

$$6. \int_0^{\pi/2} \int_0^{2x} \int_0^{x+z} \cos(x-2y+z) \, dy \, dz \, dx$$

$$\Rightarrow \int_0^{\pi/2} \int_0^{2x} \int_0^{x+z} \cos(x-2y+z) \, dy \, dz \, dx$$

$$\Rightarrow \int_0^{\pi/2} \int_0^{2x} \left[\frac{\sin(x-2y+z)}{2} \right]_0^{x+z} dz \, dx$$

$$\Rightarrow \frac{1}{2} \int_0^{\pi/2} \int_0^{2x} \left[\sin(x-2(x+z)+z) \right] - \left[\sin(x-2(0)+z) \right] dz \, dx$$

$$\Rightarrow \frac{1}{2} \int_0^{\pi/2} \int_0^{2x} \sin(x-2x-2z+z) - \sin(x+z) \, dz \, dx$$

$$\Rightarrow \frac{1}{2} \int_0^{\pi/2} \int_0^{2x} \sin(-x-z) - \sin(x+z) \, dz \, dx$$

$$\Rightarrow \frac{1}{2} \int_0^{\pi/2} \int_0^{2x} -\sin(x+z) - \sin(x+z) \, dz \, dx$$

$$\Rightarrow \frac{1}{2} \int_0^{\pi/2} \int_0^{2x} -2\sin(x+z) \, dz \, dx$$

$$\Rightarrow \int_0^1 [2(2y)^3 y + 3(4y^2 y^2) - (2(y)^3 y + 3(y)^4)] dy$$

$$\Rightarrow \int_0^1 [(16y^4 + 12y^4) - (2y^4 + 3y^4)] dy$$

$$\Rightarrow \int_0^1 16y^4 + 12y^4 - 2y^4 - 3y^4 dy$$

$$\Rightarrow \int_0^1 23y^4 dy$$

$$\Rightarrow \left[\frac{23y^5}{5} \right]_0^1$$

$$= \frac{23(1)^5}{5} - \frac{23(0)^5}{5}$$

$$= \boxed{\frac{23}{5}} \text{ Ans.}$$

$$5. \int_1^2 \int_0^{2z} \int_0^{\ln x} x e^{-y} dy dx dz$$

$$\Rightarrow \int_1^2 \int_0^{2z} \int_0^{\ln x} x e^{-y} dy dx dz$$

$$\Rightarrow \int_1^2 \int_0^{2z} \left[\frac{x e^{-y}}{(-1)} \right]_0^{\ln x} dx dz$$

$$\Rightarrow \int_1^2 \int_0^{2z} (-x e^{-(\ln x)}) - (-x e^{-(0)}) dx dz$$

$$\Rightarrow \int_1^2 \int_0^{2z} -x e^{-\ln x} + x dx dz$$

$$\Rightarrow \int_1^2 \int_0^{2z} -x e^{-\ln x} + x dx dz$$

$$\Rightarrow \int_1^2 \int_0^{2z} \frac{-x}{x} + x dx dz$$

$$\Rightarrow \int_1^2 \int_0^{2z} -1 + x dx dz$$

$$\Rightarrow \int_1^2 \left[-x + \frac{x^2}{2} \right]_0^{2z} dz$$

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$$\Rightarrow -\frac{16}{3} + \frac{32}{5} - \frac{-80+96}{15} = \boxed{\frac{16}{15}}$$

$$4. \iiint_0^1 \int_0^{2y} \int_0^{x+y} 6xy \, dz \, dx \, dy \quad \text{Ans}$$

$$\Rightarrow \int_0^1 \int_y^{2y} \int_0^{x+y} 6xy \, dz \, dx \, dy$$

$$\Rightarrow \int_0^1 \int_y^{2y} [6xy z]_0^{(x+y)} \, dx \, dy$$

$$\Rightarrow \int_0^1 \int_y^{2y} [6xy(x+y)] - 0 \, dx \, dy$$

$$\Rightarrow \int_0^1 \int_y^{2y} (6x^2y + 6xy^2) \, dx \, dy$$

$$\Rightarrow \int_0^1 \left[\frac{26x^3y}{3} + \frac{36x^2y^2}{2} \right]_y^{2y} \, dy$$

$$\Rightarrow \int_0^1 (2x^3y + 3x^2y^2) \Big|_y^{2y} \, dy$$

Derivative Properties:-

$$\mathcal{L}\{f'(t)\} = s\mathcal{L}\{f(t)\} - f(0) \quad \text{OR}$$

$$\mathcal{L}\{f'(t)\} = sF(s) - f(0)$$

$$\mathcal{L}\{f''(t)\} = s^2\mathcal{L}\{f(t)\} - sf(0) - f'(0) \quad \text{OR}$$

$$\mathcal{L}\{f''(t)\} = s^2F(s) - sf(0) - f'(0)$$

$$\mathcal{L}\{f'''(t)\} = s^3\mathcal{L}\{f(t)\} - s^2f(0) - sf'(0) - f''(0) \quad \text{OR}$$

$$\mathcal{L}\{f'''(t)\} = s^3F(s) - s^2f(0) - sf'(0) - f''(0)$$

Second Shift Theorem:-

$$\mathcal{L}\{f(t-a)\mathcal{U}(t-a)\} = e^{-as}\mathcal{L}\{f(t)\}$$

Example 4:- Evaluate $\mathcal{L}\{(t-3)^2\mathcal{U}(t-3)\}$

Sol:-

Here $f(t) = t^2$, so

$$F(s) = \mathcal{L}\{f(t)\} = \mathcal{L}\{t^2\} = \frac{2}{s^3}, s > 0$$

By second translation theorem, we have

$$e^{at} = \frac{1}{s-a}$$

$$(ix) \mathcal{L}\{\sinh kt\} = \textcircled{c}$$

$$(x) \mathcal{L}\{\cosh kt\} = \textcircled{d}$$

$$(xi) \frac{1}{3} \mathcal{L}\{\sin 3t\} = \textcircled{a}$$

$$(xii) \mathcal{L}\{2+t\} = \textcircled{a}$$

$$(xiii) \mathcal{L}\{e^{-3t}\} = e^{(a)t} = \frac{1}{s} + \frac{1}{3}$$

$$(xiv) \mathcal{L}\{e^{2t} + e^{3t}\} = \textcircled{b}$$

$$\frac{1}{s-2} + \frac{1}{s-3}$$

$$\frac{(s-3) + (s-2)}{(s-2)(s-3)} = \frac{s-3+s-2}{(s-2)(s-3)} =$$

$$\frac{2s-5}{(s-2)(s-3)}$$

$\textcircled{d}$

$$(xv) \mathcal{L}\{e^{2t} - e^{3t}\} =$$

$$\frac{1}{s-2} - \frac{1}{s-3} = \frac{(s-3) - (s-2)}{(s-2)(s-3)}$$

$$6. \mathcal{L}\{\sinh kt\} = \frac{k}{s^2 - k^2}$$

$$7. \mathcal{L}\{\cosh kt\} = \frac{s}{s^2 - k^2}$$

MCQs:-

$$(i) \mathcal{L}\{f(t)\} = \textcircled{d}$$

$$(ii) s \mathcal{L}\{1\} = \textcircled{b}$$

$$(iii) \mathcal{L}\{t\} = \textcircled{c}$$

$$(iv) \mathcal{L}\{t\} = \textcircled{d}$$

$$(v) \mathcal{L}\{t^n\} = \textcircled{d}$$

$$(vi) \mathcal{L}\{e^{at}\} = \textcircled{b}$$

$$(vii) \mathcal{L}\{\sin kt\} = \textcircled{a}$$

$$(viii) \mathcal{L}\{\cos kt\} = \textcircled{b}$$

$$= \left[-\frac{\sin t e^{-st}}{s} \right]_0^\pi + \frac{1}{s} \cos t e^{-st} \Big|_0^\pi - \int_0^\pi \frac{d(\cos t)}{dt} \cdot e^{-st} dt$$

$$= \left[-\frac{\sin t e^{-st}}{s} \right]_0^\pi + \left(\frac{1}{s} \left[\cos t e^{-st} \right]_0^\pi - \int_0^\pi \frac{-\sin t e^{-st}}{-s} dt \right)$$

$$= \left[-\frac{\sin t e^{-st}}{s} \right]_0^\pi + \left(\frac{1}{s^2} \cos t e^{-st} \right) \Big|_0^\pi + \int_0^\pi \frac{\sin t e^{-st}}{-s} dt$$

$$= 0 \left[-\frac{1}{s^2} \cos t e^{-st} \right]_0^\pi + \int_0^\pi \sin t e^{-st} dt$$

$$= \left[-\frac{1}{s^2} \cos t e^{-st} \right]_0^\pi + \mathcal{L}\{f(t)\}$$

$$= -\frac{1}{s^2} \left[\cos t e^{-st} \right]_0^\pi + \mathcal{L}\{f(t)\}$$

$$\mathcal{L}\{f(t)\} = -\frac{1}{s^2} \left[\frac{-1}{e^{s(\pi)}} - \frac{-1}{e^{s(0)}} \right] + \mathcal{L}\{f(t)\}$$

$$= \frac{1}{s^2 e^{s\pi}} + \frac{1}{s^2} + \frac{1}{s^2} \mathcal{L}\{f(t)\}$$

$$\mathcal{L}\{f(t)\} + \frac{1}{s^2} \mathcal{L}\{f(t)\} = \frac{1}{s^2 e^{s\pi}} + \frac{1}{s^2}$$

$$= \frac{s^2 \mathcal{L}\{f(t)\} + \mathcal{L}\{f(t)\}}{s^2} = \frac{1 + e^{-s\pi}}{s^2}$$

$$= \mathcal{L}\{f(t)\} \left(\frac{s^2 + 1}{s^2} \right) = \frac{1 + e^{-s\pi}}{s^2}$$

$$\boxed{\mathcal{L}\{f(t)\} = \frac{1 + e^{-s\pi}}{s^2 + 1}} \quad \text{Ans}$$

Formulas :-

$$1. \mathcal{L}\{1\} = \frac{1}{s}$$

$$2. \mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}, \quad n = 1, 2, \dots$$

$$3. \mathcal{L}\{e^{at}\} = \frac{1}{s-a}$$

$$4. \mathcal{L}\{\sin kt\} = \frac{k}{s^2 + k^2}$$

$$5. \mathcal{L}\{\cos kt\} = \frac{s}{s^2 + k^2}$$

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$$= -\frac{1}{s^2} - \frac{1}{s} \left(\frac{1}{e^s} - \frac{1}{1} \right) + \frac{1}{s e^s}$$

$$= \left[+\frac{1}{s^2} - \frac{1}{s e^s} \right] \text{ Ans}$$

Example 5:-

Show that $\mathcal{L}\{f(t)\} = \frac{e^{-3s}}{s}, s > 0$

where

$$f(t) = \begin{cases} 0, & 0 \leq t \leq 3 \\ 1, & t > 3 \end{cases}$$

Sol

$$\mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} f(t) dt$$

$$\mathcal{L}\{f(t)\} = \int_0^3 e^{-st} \cdot 0 dt + \int_3^{\infty} e^{-st} \cdot 1 dt$$

$$= 0 + \left[\frac{e^{-st}}{-s} \right]_3^{\infty}$$

$$= -\frac{1}{s} \left[\frac{1}{e^{s(\infty)}} - \frac{1}{e^{s(3)}} \right] = \frac{1}{s} \left(\frac{1}{e^{3s}} \right) \quad s > 0$$

Ans

Example 8:-

Find the Laplace transform of

$$f(t) = \begin{cases} \sin t, & 0 \leq t \leq \pi \\ 0, & t > \pi \end{cases}$$

Sol

formula

$$\mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} f(t) dt$$

$$\mathcal{L}\{f(t)\} = \int_0^{\pi} e^{-st} \sin t dt + \int_{\pi}^{\infty} e^{-st} \cdot 0 dt$$

$$= \sin t \int_0^{\pi} e^{-st} dt - \int_0^{\pi} \left[\frac{d(\sin t)}{dt} \right] \cdot \int_0^{\pi} e^{-st} dt dt + 0$$

$$= \left[-\frac{\sin t}{s} e^{-st} \right]_0^{\pi} - \int_0^{\pi} \cos t \cdot \left[\frac{e^{-st}}{-s} \right] dt$$

$$= \left[-\frac{\sin t}{s} e^{-st} \right]_0^{\pi} + \frac{1}{s} \int_0^{\pi} \cos t \cdot e^{-st} dt$$

$$0 \leq f(x) < \infty$$

Date: 9/may/26:-

Q Find the Laplace Transform of the function

$$f(t) = \begin{cases} t, & 0 \leq t < 1 \\ 1, & t \geq 1 \end{cases}$$

Sol
formula

$$\mathcal{L}[f(t)] = \int_0^{\infty} e^{-st} f(t) dt$$

$$\mathcal{L}[f(t)] = \int_0^1 e^{-st} f(t) dt + \int_1^{\infty} e^{-st} f(t) dt$$

$$= \int_0^1 e^{-st} t dt + \int_1^{\infty} e^{-st} 1 dt$$

$$= \int_0^1 t e^{-st} dt + \left[\frac{e^{-st}}{-s} \right]_1^{\infty}$$

$$= t \int_0^1 e^{-st} dt - \int_0^1 \left(\frac{d}{dt} t \right) \cdot \int_0^1 e^{-st} dt dt + \left[\frac{e^{-st}}{-s} \right]_1^{\infty}$$

$$= \frac{t e^{-st}}{-s} - \int_0^1 \left[\frac{e^{-st}}{-s} \right] dt + \left[\frac{e^{-st}}{-s} \right]_1^{\infty}$$

$$= \frac{t e^{-st}}{-s} - \left(\frac{1}{s} \right) \left[\frac{e^{-st}}{-s} \right] dt + \left(\frac{e^{-st}}{-s} \right)$$

$$= \frac{t e^{-st}}{-s} + \frac{1}{s} \left[\frac{e^{-st}}{-s} \right]$$

$$\left[\frac{t e^{-st}}{-s} \right]_0^1 - \int_0^1 \frac{e^{-st}}{-s} dt + \left[\frac{e^{-st}}{-s} \right]_1^{\infty}$$

$$= \left[\frac{t e^{-st}}{-s} \right]_0^1 + \left[\frac{e^{-st}}{-s^2} \right]_0^1 + \left(\frac{-1}{s} \right) \left[\frac{1}{e^{s(\infty)}} - \frac{1}{e^{s(0)}} \right]$$

$$= \frac{e^{-s}}{-s} + \left[\frac{e^{-st}}{-s^2} \right]_0^1 - \frac{1}{s} \left[0 - e^{-s} \right]$$

$$= \frac{e^{-s}}{-s} + \frac{1}{s^2} \left[\frac{1}{e^{s(1)}} - \frac{1}{e^{s(0)}} \right] - \frac{1}{s} \left[\frac{1}{e^{s(\infty)}} - \frac{1}{e^{s(0)}} \right]$$

Jun 6, 2026, 6:00 PM

Q. No 2 Find the Laplace transform of t^n

formula: $\mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} f(t) dt$

So we have to find

$$\mathcal{L}\{t^n\} = \mathcal{L}\{t^n\} = \int_0^{\infty} e^{-st} t^n dt$$

$$= \int_0^{\infty} e^{-st} t^n dt$$

$$\Rightarrow t^n \int_0^{\infty} e^{-st} \left[\frac{d(t^n)}{dt} \int_0^{\infty} e^{-st} dt \right] dt$$

$$\Rightarrow \frac{t^n e^{-st}}{-s} \Big|_0^{\infty} - \int_0^{\infty} n t^{n-1} \frac{e^{-st}}{-s} dt$$

$$\Rightarrow \frac{t^n e^{-st}}{s} \Big|_0^{\infty} - \int_0^{\infty} \frac{(n t^{n-1}) e^{-st}}{s} dt$$

$$\Rightarrow \frac{t^n}{s e^{st}} \Big|_0^{\infty} + \frac{n}{s} \int_0^{\infty} n t^{n-1} e^{-st} dt$$

$$\Rightarrow \frac{1}{s} \left[\lim_{t \rightarrow \infty} \frac{t^n}{e^{st}} - \frac{0^n}{e^{0(0)}} \right] + \frac{n}{s} \mathcal{L}\{t^{n-1}\}$$

$$= \frac{1}{s} [0 - 0] + \frac{n}{s} \mathcal{L}\{t^{n-1}\}$$

$$\mathcal{L}\{t^n\} = \frac{n}{s} \cdot \mathcal{L}\{t^{n-1}\} = \frac{n}{s} \cdot \frac{(n-1)}{s} \mathcal{L}\{t^{n-2}\}$$

$$\mathcal{L}\{t^n\} = \frac{n(n-1) \dots (2)}{s^{n-1}} \mathcal{L}\{1\}$$

$$= \frac{n(n-1)(n-2) \dots 2 \cdot 1}{s^{n-1}} \cdot \frac{1}{s} = \frac{n!}{s^n}$$

$$\mathcal{L}\{t^n\} = \frac{n}{s} \mathcal{L}\{t^{n-1}\}$$

$$= \frac{n}{s} \cdot \frac{(n-1)}{s} \mathcal{L}\{t^{n-2}\}$$

$$= \frac{n(n-1)(n-2)}{s^3} \mathcal{L}\{t^{n-3}\}$$

$$= \frac{n(n-1)(n-2) \dots 2 \cdot 1}{s^n} \mathcal{L}\{t^0\}$$

$$= \frac{n!}{s^n} \cdot \frac{1}{s}$$

$$= \frac{n!}{s^{n+1}} \quad \text{Ans}$$

$$\mathcal{L}[1] = \int_0^{\infty} e^{-st} \cdot 1 \, dt$$

$$= \left. \frac{e^{-st}}{-s} \right|_0^{\infty}$$

$$= -\frac{1}{s} \cdot \left. \frac{1}{e^{st}} \right|_0^{\infty}$$

$$= -\frac{1}{s} \left[\frac{1}{e^{s(\infty)}} - \frac{1}{e^{s(0)}} \right]$$

$$= -\frac{1}{s} \left[\frac{1}{\infty} - \frac{1}{1} \right]$$

$$= -\frac{1}{s} [0 - 1]$$

$$e^{\infty} = \infty$$

$$\frac{1}{\infty} = 0$$

$$= \frac{1}{s} \text{ Ans}$$

$$Ex = 15.6$$

QNO 31 to QNO 32

QNO 28 to QNO 30

Date: 2/may/26

Laplace Transform

If a function $f(t)$ is defined in the region

$0 \leq t < \infty$ where t are real, then the integral

$$\int_0^{\infty} e^{-st} f(t) dt$$

is called Laplace transformation.

8QNO1 find the Laplace transform of 1.

Solution: Here $f(t) = 1$

We have to find $\mathcal{L}[f(t)]$

$$\mathcal{L}[f(t)] = \int_0^{\infty} e^{-st} f(t) dt$$